EXPERIMENT 1: Customer Churn Predictor using Kaggle Telco Dataset

AIM:

To predict customer churn using the Kaggle Telco Customer Churn dataset by preprocessing the data, applying One-Hot Encoding and StandardScaler, training a Logistic Regression model, and evaluating it using Accuracy, Confusion Matrix, Classification Report, and ROC-AUC.

PROCEDURE:

1. Loadthe Telco Customer Churn dataset directly from the Kaggle/IBM raw CSV URL using Pandas.
2. Inspect the dataset and display its shape and sample records.
3. Remove the unnecessary customerID column.
4. Convert the TotalCharges column into numeric format and treat blank values as missing values.
5. Remove rows containing missing values using dropna().
6. Convert the target variable Churn from Yes/No into 1/0.
7. Separate the dataset into input features X and target variable y.
8. Apply One-Hot Encoding to categorical features using pd.get_dummies().
9. Split the dataset into 80% training and 20% testing data using train_test_split() with stratified sampling.
10. Apply StandardScaler to scale the training and testing features.
11. Train a Logistic Regression model using the scaled training data.
12. Predict customer churn for the test data and calculate churn probabilities.
13. Evaluate the model using Accuracy, Confusion Matrix, Classification Report, and ROC-AUC Score.
14. Display the important features influencing customer churn using the Logistic Regression coefficients.

PROGRAM:

```
# ============================================================
# Experiment 1: Customer Churn Predictor
# Dataset: Telco Customer Churn
# ============================================================

import pandas as pd
import numpy as np

from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import (
    accuracy_score,
    confusion_matrix,
    classification_report,
    roc_auc_score
)

#Step 1: Load Dataset
DATA_URL = "https://raw.githubusercontent.com/IBM/telco-customer-churn-on-icp4d/master/data/Telco-Customer-Churn.csv"

df = pd.read_csv(DATA_URL)

print("--- Dataset Loaded Successfully ---")
print("Dataset Shape:", df.shape)
print(df[['customerID', 'tenure',
        'MonthlyCharges', 'TotalCharges', 'Churn']].head())
```

```
# Step 2: Data Cleaning
df.drop(columns=['customerID'], inplace=True)

df['TotalCharges'] = pd.to_numeric(
   df['TotalCharges'],
   errors='coerce'
)

df.dropna(inplace=True)

# Step 3: Encode Target
df['Churn'] = df['Churn'].map({
   'Yes': 1,
   'No': 0
})

#Step 4: Separate Features and Target
X= df.drop(columns=['Churn'])
y= df['Churn']

#Step 5: One-Hot Encoding
X_encoded = pd.get_dummies(
   X,
   drop_first=True
)

#Step 6: Train-Test Split
X_train, X_test, y_train, y_test = train_test_split(
   X_encoded,
   y,
   test_size=0.20,
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    random_state=42,
    stratify=y
)

#Step 7: Feature Scaling
scaler = StandardScaler()

X_train_scaled = scaler.fit_transform(X_train)
X_test_scaled = scaler.transform(X_test)

# Step 8: Train Logistic Regression Model
model = LogisticRegression(max_iter=1000)

model.fit(X_train_scaled, y_train)

# Step 9: Prediction
y_pred = model.predict(X_test_scaled)

y_proba = model.predict_proba(
    X_test_scaled
)[:, 1]

# Step 10: Evaluation
accuracy = accuracy_score(y_test, y_pred)
roc_auc = roc_auc_score(y_test, y_proba)
cm = confusion_matrix(y_test, y_pred)

print("\n--- Model Performance ---")
print(f"Accuracy: {accuracy * 100:.2f}%")
print(f"ROC-AUC: {roc_auc:.4f}")
```

```
print("\nConfusion Matrix:")
print(cm)

print("\nClassification Report:")
print(classification_report(y_test, y_pred))

# Step 11: Important Churn Factors
feature_weights = pd.Series(
    model.coef_[0],
    index=X_encoded.columns
)

print("\n--- Top 3 Factors Increasing Churn ---")
print(feature_weights.sort_values(
    ascending=False
).head(3))

print("\n--- Top 3 Factors Reducing Churn ---")
print(feature_weights.sort_values().head(3))
```

OUTPUT:

```
--- Dataset Loaded Successfully ---
Dataset Shape: (7043, 21)
   customerID  tenure  MonthlyCharges TotalCharges Churn
0  7590-VHVEG       1           29.85        29.85    No
1  5575-GNVDE      34           56.95       1889.5    No
2  3668-QPYBK       2           53.85       108.15   Yes
3  7795-CFOCW      45           42.30      1840.75    No
4  9237-HQITU       2           70.70       151.65   Yes

--- Model Performance ---
Accuracy: 80.38%
ROC-AUC: 0.8357

Confusion Matrix:
[[916 117]
 [159 215]]

Classification Report:
              precision    recall  f1-score   support

           0       0.85      0.89      0.87      1033
           1       0.65      0.57      0.61       374

    accuracy                           0.80      1407
   macro avg       0.75      0.73      0.74      1407
weighted avg       0.80      0.80      0.80      1407


--- Top 3 Factors Increasing Churn ---
InternetService_Fiber optic    0.727745
TotalCharges                   0.639028
StreamingTV_Yes                0.249702
dtype: float64

--- Top 3 Factors Reducing Churn ---
tenure              -1.347613
MonthlyCharges      -0.851551
Contract_Two year   -0.602591
dtype: float64
```

RESULT:

Thus, the Customer Churn Prediction model was successfully developed using the Telco Customer Churn dataset. The data was cleaned and pre-processed using One-Hot Encoding and StandardScaler, and Logistic Regression was used to predict customer churn. The model was successfully evaluated using Accuracy, Confusion Matrix, Classification Report, and ROC-AUC.